

03 — Open Science Plan

TriadicFrameworks: An Open-Science Substrate for Regime-Aware Modeling, Scanning, and Simulation

Submitted to NASA High Priority Open-Source Science (HPOSS)

1. Commitment to Open Science

TriadicFrameworks is designed from the outset as an open-science research infrastructure. All protocols, specifications, models, and reference implementations produced under this project will be openly licensed, publicly accessible, and archived with persistent identifiers to ensure long-term availability and reproducibility.

The project aligns fully with NASA's Open-Source Science Initiative (OSSI) by prioritizing transparency, community accessibility, and reproducible scientific workflows.

2. Open Licensing and Public Repositories

All software, documentation, and specifications will be released under a permissive open-source license (MIT or Apache 2.0). Public repositories will be maintained on GitHub under the TriadicFrameworks organization, with versioned releases archived on Zenodo and assigned DOIs.

Deliverables include:

- dsrsp/0.1 specification and schemas
- RSM and vST alignment layer documentation
- reference implementations and libraries
- ILP module
- example datasets and workflows
- integration guides and developer documentation

All materials will remain freely accessible without registration or usage restrictions.

3. Reproducibility and Transparency

The project will provide reproducible workflows, including:

- example sensor envelopes and regime profiles
- Jupyter notebooks demonstrating classification and validation
- simulation-based examples for testing dsrsp/0.1
- step-by-step integration guides for RSM and vST engines

Each workflow will include:

- clear instructions
- versioned dependencies
- open datasets or synthetic data generation scripts

This ensures that researchers can reproduce results, validate behavior, and extend the framework for their own scientific applications.

4. Data and Software Management

All data products generated during the project will be:

- openly licensed
- documented with metadata
- archived in public repositories
- versioned and assigned DOIs when appropriate

Software deliverables will follow open-source best practices:

- semantic versioning
- continuous integration
- issue tracking
- community contribution guidelines
- automated documentation builds

This ensures long-term sustainability and community trust.

5. Community Engagement and Accessibility

TriadicFrameworks will support community adoption through:

- public documentation and tutorials
- example integrations for sensing and simulation systems
- developer-focused onboarding materials
- a public discussion forum or GitHub Discussions space
- periodic updates and release notes

The project will also provide clear pathways for community contributions, including code, documentation, and scientific extensions.

6. Long-Term Sustainability

By publishing all components under permissive licenses, archiving releases with DOIs, and maintaining public repositories, TriadicFrameworks ensures that the framework remains accessible and extensible beyond the performance period.

The project's modular design allows researchers, developers, and institutions to adopt individual components (dsrsp/0.1, RSM, vST, SLRP) or integrate the full substrate into their scientific workflows.

7. Summary

TriadicFrameworks is inherently aligned with NASA's open-science mission. The project will deliver openly licensed protocols, models, and tools that support transparent, reproducible, and community-accessible scientific analysis across remote sensing, planetary exploration, and simulation environments.

All deliverables will be publicly available, versioned, documented, and archived to ensure long-term scientific value and broad community impact.